Computer Vision and Image Processing Project (2025/26)

Deep Learning for Traffic Sign Recognition using the GTSRB Dataset

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***Abstract*—This report presents the first phase of a deep learning project focused on traffic sign recognition using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. The main objective of this phase was to study the impact of different data augmentation techniques and image preprocessing strategies on the performance of convolutional neural networks (CNNs). Several augmentation configurations were evaluated, including geometric transformations, color-based augmentations, combined augmentations, and massive dataset expansion. Experimental results demonstrate that data augmentation significantly improves model generalization and robustness, achieving a maximum average test accuracy of approximately 98.38%.**

***Index Terms*—Deep Learning, Computer Vision, Traffic Sign Recognition, GTSRB, Convolutional Neural Networks, Data Augmentation, Image Processing, Ensemble Learning**

I. Introduction

Traffic sign recognition is an important task in computer vision and autonomous driving systems. Modern Advanced Driver Assistance Systems (ADAS) rely heavily on robust visual recognition models capable of detecting and classifying traffic signs under different environmental conditions such as illumination changes, rotations, occlusions, and motion blur.

The objective of this project is to explore deep learning techniques applied to the GTSRB dataset in order to achieve the highest possible classification accuracy. According to published benchmarks, the best reported accuracy on the dataset is approximately 99.82%.

This first phase focuses mainly on:

- Studying image preprocessing techniques;
- Applying static and dynamic data augmentation;
- Training multiple CNN models with different augmentation pipelines;
- Comparing the impact of augmentation strategies on model performance.

II. Dataset

The German Traffic Sign Recognition Benchmark (GTSRB) is a well-known benchmark dataset for traffic sign classification.

The dataset contains:

- 43 traffic sign classes;
- More than 50,000 images;
- Images captured under real-world conditions;
- Significant variability in illumination, perspective, scale, and occlusion.

The dataset was divided into:

- Training set;
- Validation set;
- Test set.

A. Class Distribution

One important characteristic of the dataset is the imbalance between classes. Some traffic signs contain considerably more samples than others, which can introduce bias during training.

The initial dataset analysis revealed:

- Strong variability in the number of samples per class;
- Variability in image resolution;
- Presence of noisy samples and illumination changes.

III. Methodology

A. Framework and Libraries

The project was implemented using:

- Python;
- PyTorch;
- Torchvision;
- NumPy;
- Matplotlib;
- Pandas.

B. Image Preprocessing

Before training, all images were resized to a fixed dimension in order to ensure consistency during batch processing.

The preprocessing pipeline included:

- Image resizing;
- Conversion to tensor format;
- Normalization;
- Data type conversion to floating point.

Example preprocessing pipeline:

```
transform_base = v2.Compose([
    v2.Resize((IMG_SIZE, IMG_SIZE)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])
```

IV. Model Architecture

A convolutional neural network (CNN) architecture was implemented for traffic sign classification.

The network includes:

- Convolutional layers;
- Batch normalization;
- ReLU activation functions;
- Max pooling layers;
- Fully connected layers;
- Dropout regularization.

The model was designed to balance:

- Computational efficiency;
- Generalization capability;
- Classification accuracy.

V. Data Augmentation Strategies

Data augmentation is a key technique to improve the generalization ability of deep learning models, especially in image classification tasks such as traffic sign recognition. In this work, multiple augmentation strategies were systematically designed and evaluated using the GTSRB dataset.

The main objective was to analyze how different transformations affect model robustness and accuracy under real-world variations such as rotation, illumination changes, occlusions, and perspective distortion.

All experiments follow a controlled evaluation protocol using the same CNN architecture, optimizer (Adam), and training configuration, varying only the augmentation pipeline.

A. Experiment 0 - Baseline

The baseline experiment uses only minimal preprocessing, without any augmentation applied to the training data. This serves as the reference for all subsequent comparisons.

```
results_base, mean_base, std_base = run_experiment(
    name='baseline',
    transform_train=transform_base,
    transform_test=transform_base
)
```

B. Experiment 1 - Geometric Augmentation

This experiment introduces basic geometric transformations, aiming to improve invariance to small spatial changes such as rotation and translation.

```
transform_geo = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
    InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE, IMG_SIZE)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])
```

These transformations simulate small camera shifts and orientation changes commonly found in real driving scenarios.

C. Experiment 2 - Advanced Geometric Augmentation

A more aggressive geometric pipeline was introduced using perspective distortion and random cropping. This aims to simulate more extreme viewpoint variations.

```
transform_geo2 = v2.Compose([
    v2.RandomPerspective(distortion_scale=0.2, p=0.6),
    v2.Resize((IMG_SIZE + 8, IMG_SIZE + 8)),
    v2.RandomCrop((IMG_SIZE, IMG_SIZE)),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])
```

This setup introduces stronger spatial variability while maintaining the semantic structure of traffic signs.

D. Experiment 3 - Color Augmentation

This experiment focuses on photometric transformations to improve robustness to lighting conditions, shadows, and weather variations.

```
transform_color = v2.Compose([
    v2.Resize((IMG_SIZE, IMG_SIZE)),
    v2.ColorJitter(
        brightness=0.5,
        contrast=0.5,
        saturation=0.3,
        hue=0.05
    ),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True)
])
```

Color jittering is particularly relevant for real-world driving scenarios where illumination is highly variable.

E. Experiment 4 - Combined Augmentation

This experiment combines geometric and color augmentations, along with random erasing to simulate occlusions such as dirt, shadows, or partial obstruction.

```
transform_combined = v2.Compose([
    v2.RandomRotation(15, interpolation=v2.
    InterpolationMode.BILINEAR),
    v2.Resize((IMG_SIZE, IMG_SIZE)),
    v2.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    v2.ColorJitter(
        brightness=0.5,
        contrast=0.5,
        saturation=0.3,
        hue=0.05
    ),
    v2.ToImage(),
    v2.ToDtype(torch.float32, scale=True),
    v2.RandomErasing(
        p=0.3,
        scale=(0.02, 0.1),
        value='random'
    )
])
```

Random erasing improves robustness by forcing the network to learn partial-feature representations.

F. Experiment 5 - Massive Dataset Expansion

This experiment significantly increases dataset diversity by duplicating the training dataset with a strongly augmented version. The final dataset is a concatenation of original and augmented samples.

```
full_original  = torchvision.datasets.ImageFolder(
    root=PATH_TRAIN,
    transform=transform_base
)

full_augmented = torchvision.datasets.ImageFolder(
    root=PATH_TRAIN,
    transform=transform_strong
)

expanded_dataset = torch.utils.data.ConcatDataset(
    [full_original, full_augmented]
)
```

This strategy effectively doubles the dataset size, improving generalization capacity.

The model is trained multiple times with different random initializations to ensure statistical reliability:

```
for i in range(RUNS):
    model = Conv_II(NUM_CLASSES).to(device)
    ...
    train(...)
    acc = evaluate(model, test_loader)
```

This setup also includes:

- Learning rate scheduling (ReduceLROnPlateau);
- Early stopping;
- Model checkpointing.

G. Experiment 6 - MixUp Augmentation

This experiment uses MixUp, a data augmentation technique that generates synthetic training samples by interpolating pairs of images and their labels.

```
mixup = v2.MixUp(num_classes=NUM_CLASSES)

inputs, targets_mix = mixup(inputs, targets)
outputs = model(inputs)
loss = loss_fn(outputs, targets_mix)
```

Unlike standard training, MixUp produces soft labels, improving decision boundaries and reducing overfitting.

During evaluation, standard validation is performed without MixUp to ensure fair performance measurement.

VI. Experimental Results TODO

VII. Discussion

The experiments demonstrate several important observations:

- The baseline model already achieved strong performance due to the quality of the dataset and the CNN architecture.
- Geometric augmentation did not significantly improve performance, possibly because excessive transformations introduced unrealistic samples.
- Color augmentation slightly improved robustness while maintaining stable results.
- Combined augmentation introduced additional variability but may have increased training difficulty.
- Massive dataset expansion achieved the best overall performance, indicating that stronger regularization and diversity improved generalization.

The best average test accuracy obtained during Phase 1 was:

$$98.38\%$$

Although this result is below the state-of-the-art benchmark (99.82%), it demonstrates that augmentation strategies positively impact model robustness and generalization.

VIII. Conclusion

This first phase successfully explored the application of deep learning techniques to the GTSRB traffic sign dataset.

The experiments demonstrated that:

- Data augmentation is essential for improving generalization;
- Different augmentation techniques have distinct impacts on performance;
- Stronger augmentation and dataset expansion yielded the best results.

The best model achieved approximately 98.38% accuracy on the test set, providing a strong baseline for future improvements.

The second phase of the project will focus on ensemble learning techniques by combining the best-performing neural networks trained during Phase 1. The objective is to evaluate whether ensembles can improve classification accuracy, robustness, and generalization performance on the GTSRB dataset, potentially approaching state-of-the-art results.

References

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